

LESSON 5-3 • SESSION 1 • GUIDED NOTES**Trapezoids by Decomposing**

Learning target – I can find the area of a trapezoid by decomposing it into a rectangle and triangles. · 6.G.A.1

WORDS TO KNOW*fill in the missing words*

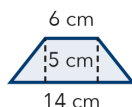
WORD	WHAT IT MEANS	EXAMPLE
trapezoid <i>trapezio</i>	A _____ with one pair of parallel sides (the bases).	bases 6 cm and 14 cm
height <i>altura</i>	The _____ distance between the two bases.	h, at a square corner
decompose <i>descomponer</i>	To _____ a shape into smaller shapes you know.	rectangle + 2 triangles

KEY IDEA*complete each sentence*

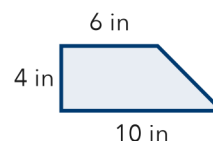
- Cut a trapezoid into a _____ and one or two triangles.
- Find the area of each piece, then _____ the areas.
- Together, the triangle bases equal the _____ of the two bases.

WORKED EXAMPLE*fill in the blanks as we solve*

An isosceles trapezoid has bases 6 cm and 14 cm and a height of 5 cm. Find its area.

**Step 1** Rectangle: $6 \times 5 = \underline{\hspace{1cm}}$ cm².**Step 2** Leftover: $14 - 6 = 8$, so each triangle base is _____ cm.**Step 3** Both triangles: $2 \times (\frac{1}{2} \times 4 \times 5) = \underline{\hspace{1cm}}$ cm².**Answer** Area = $30 + 20 = \underline{\hspace{1cm}}$ cm²**YOU TRY***use the example as a model*

1. This trapezoid has one square corner. Decompose it and find its area.

*Show your work.***Answer** _____ in²

2. Each trapezoid has two matching end triangles. Find the base of one triangle.

a. bases 8 m and 14 m _____**b.** bases 12 cm and 20 cm _____**SUM IT UP***in your own words*

To find a trapezoid's area by decomposing, I _____ and then _____.

LESSON 5-3 • SESSION 2 • GUIDED NOTES

The Trapezoid Area Formula

Learning target – I can find the area of a trapezoid using the trapezoid area formula. · 6.G.A.1

WORDS TO KNOW

fill in the missing words

WORD	WHAT IT MEANS	EXAMPLE
base <i>base</i>	One of the two _____ sides of a trapezoid.	b1 and b2
height <i>altura</i>	The _____ distance between the two bases.	h
formula <i>fórmula</i>	A _____ written with letters and numbers.	$A = \frac{1}{2}(b1 + b2)h$
right trapezoid <i>trapezio rectángulo</i>	A trapezoid with two _____ angles.	its straight side is the height

KEY IDEA

complete each sentence

- Area of a trapezoid: $A = \frac{1}{2} \times (\text{_____}) \times h$.
- _____ the bases first, then multiply by the height, then take half.
- Two copies of a trapezoid make a _____, so the trapezoid is half of it.

WORKED EXAMPLE

fill in the blanks as we solve

A trapezoid has bases 6 m and 12 m and a height of 4 m. Find its area.

Step 1 Add the bases: $6 + 12 = \underline{\hspace{1cm}}$.

Step 2 Multiply by the height: $18 \times 4 = \underline{\hspace{1cm}}$.

Step 3 Take half: $72 \div 2 = \underline{\hspace{1cm}}$.

Answer $A = \underline{\hspace{1cm}} \text{ m}^2$

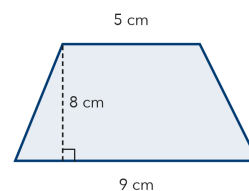
YOU TRY

use the example as a model

1. Find the area of the trapezoid.

Show your work.

Answer _____ cm^2



2. Find each area.

a. 3 m, 7 m; $h = 4 \text{ m}$ _____ m^2

b. 10 in, 12 in; $h = 5 \text{ in}$ _____ in^2

SUM IT UP

in your own words

To use the trapezoid formula, I _____ first because _____.